INCH-POUND
MIL-DTL-70438A (AR)
w/AMENDMENT 1
9 June 2011
SUPERSEDING
MIL-DTL-70438A (AR)
8 September 2008

DETAIL SPECIFICATION

Plug Assembly, Lifting, Universal

Inactive for new design after 21 August 1998

This specification is approved for use by the U.S. Army Armament Research Development and Engineering Center (ARDEC), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for the manufacturing and assembly of Universal Lifting Plugs for 155MM projectiles (see 6.1).

Comments, suggestions, or questions on this document should be addressed to the Commander, U.S. Army ARDEC, ATTN: RADR-QES-E, Picatinny, NJ 07806-5000 or emailed to ardecstdzn@conus.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 1320

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to assure the completeness of this list, document users are cautioned that they must meet all requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490 – Chemical Conversion Coatings and Pretreatments for Ferrous Surfaces (Base for Organic Coatings)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1168 – Ammunition Lot Numbering and Ammunition Data Cards

MIL-STD-1916 – DoD Preferred Methods for Acceptance of Product

(Unless otherwise indicated, Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation or contract.

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC) DRAWINGS

9345325 – Plug, Universal Assembly
9347368 – Plug, Universal
9215391 – Ternary Eutectic Alloy
9215393 – Adhesive-Sealing Compound

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(Copies of these drawings may be requested online at Drawing-Request@pica.army.mil or from U.S. Army ARDEC, ATTN: AMSRD-AAR-AIS-TD, Picatinny, NJ 07806-5000.)

2.3 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D950 - Standard Method of Test for Impact Strength of Adhesive Bonds

(Copies of these documents are available online at http://www.astm.org or from ASTM International, 100 Barr Harbor Drive, West Conshohocken, PA 19428.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Required inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance. A sample shall be subjected to conformance inspection in accordance with 4.3.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on drawing 9345325.

3.2.1 Plug, Universal. The materials, components and assemblies shall be in accordance with dwg. 9347368.

3.2.2 Universal plug assembly (prior to pouring alloy). The materials, components and assemblies shall be in accordance with dwg. 9345325.

3.2.3 Universal plug assembly (after pouring alloy). The materials, components and assemblies shall be in accordance with dwg. 9345325.

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3.2.4 Ternary eutectic alloy. The materials, components and assemblies shall be in accordance with dwg. 9215391.

3.2.5 Adhesive. The materials, components and assemblies shall be in accordance with dwg. 9215393.

3.3 Ammunition lot numbering. Ammunition lot numbers shall be assigned in accordance with MIL-STD-1168.

3.4 Tension. The plug shall sustain a load of 8000 pounds minimum without any sign of failure.

3.5 Ductility. The Plug shall withstand permanent deformation of the eye without cracking.

3.6 Fracture. The plug shall be free from porosity and shrinks or discontinuous structure.

3.7 Workmanship. All parts shall be fabricated and finished in a thorough, workmanlike manner. They shall be clean and free of burrs, sharp edges, unblended radii, cracks, surface defects, chips, dirt, grease, dust and other foreign matter. Where cleaning and preparation are required by component details, the finish specified shall apply to the indicated surfaces regardless of contour configuration. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All paint shall be dry to the touch before packing for shipment. The paint shall be considered dry when the soft tacky condition no longer exists and the paint feels firm when touched with a finger using moderate pressure.

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4. VERIFICATION

TABLE I. Requirement/Verification Cross Reference Matrix.

Methods of Verification N/A – Not Applicable 1 – Analysis 2 – Demonstration 3 – Examination 4 – Test							Classes of Verification A– First Article B– Acceptance		
Section 3 Requirements	Description	Verification Method					Verification Class		Section 4 Verification
		N/A	1	2	3	4	A	B	
3.1.1	First article				X	X	X	X	4.2
3.1.2	Conformance				X	X	X	X	4.3
3.2	Components and assemblies				X	X	X	X	4.3
3.2.1	Plug, Universal				X	X	X	X	4.3.2.1
3.2.2	Universal plug assembly (prior to pouring alloy)				X	X	X	X	4.3.2.2
3.2.3	Universal plug assembly (after pouring alloy)				X	X	X	X	4.3.2.3/4.4.2/ 4.4.5
3.2.4	Ternary eutectic alloy				X	X	X	X	4.3.2.4/4.4.1
3.2.5	Adhesive				X	X	X	X	4.3.2.5/4.4.4
3.3	Ammunition lot numbering				X		X	X	4.3.1
3.4	Tension					X	X	X	4.3.2.1/ 4.4.3.1
3.5	Ductility					X	X	X	4.3.2.1/ 4.4.3.2
3.6	Fracture					X	X	X	4.3.2.1/ 4.4.3.3
3.7	Workmanship				X		X	X	4.3.2.1/ 4.3.2.3

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4.1 Classification of verification. The verification requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection.

4.2.1 First article quantity. First article verification shall be performed on the quantity of items as indicated in table II.

4.2.1.1 Multiple cavity molds. When a multiple cavity mold is to be used in production, the sample shall include at least three plugs from each cavity in the mold. The plug shall exhibit permanent cavity identification.

4.2.2 Inspections to be performed. The first article verification shall be performed in accordance with Table II.

4.2.3 First article rejection. If any item fails to comply with requirements, the first article sample shall be rejected.

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TABLE II. FIRST ARTICLE INSPECTION

EXAMINATION OR TEST	NUMBER OF SAMPLES	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Universal Plug Dwg. 9347368 Examination for defects	Note 1	3.2.1	4.3.2.1
Universal plug assembly (prior to pouring alloy) Dwg. 9347368 Examination for defects	25	3.2.2	4.3.2.2
Universal lifting plug assembly (after pouring alloy) Dwg. 9345325 Examination for defects Salt spray test	50 2	3.2.3 3.2.3	4.3.2.3 4.4.2
Ternary eutectic alloy Dwg. 9215391 Melting test	1 pound	3.2.4	4.4.1
Adhesive Dwg. 9215393 Examination for defects	10	3.2.5	4.4.4

1. 25 minimum; at least 3 per cavity from multiple cavity molds.

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4.3 Conformance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with lot formation requirement of MIL-STD-1916 (Formation and identification of lots or batches). In addition each inspection lot shall contain ternary eutectic alloy from one lot number and plugs of one lot interfix number from one manufacturer.

4.3.2 Classification of characteristics.

a. Sampling requirements. Inspection sampling requirements for critical, major and minor characteristics are defined in MIL-STD-1916. Unless specified otherwise, Inspection Level IV shall be used for all characteristics defined as Majors and Inspection Level II for all Minor characteristics; Critical characteristics shall be addressed in accordance with MIL-STD-1916.

b. Conformance inspection. Conformance inspection shall be performed in accordance with paragraph 4.3.2.1 and 4.3.2.2. For all conformance inspections the same sample specimen may be used for all non-destructive examinations or tests unless otherwise specified.

c. Alternative Inspections. Alternative conformance inspections may be submitted and approved in accordance with MIL-STD-1916.

d. Tool control. Dimensions marked tool control shall be gaged at the beginning of production and whenever tooling is replaced. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided there is an established correlation between the tool dimensions and the component dimensions prior to the start of production. There shall be adequate surveillance to assure that the drawing requirements are being met throughout the life of the tool.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Plug, Universal	SHEET 1 OF 2	DRAWING NUMBER 9347368
			NEXT HIGHER ASSEMBLY 9345325

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major				
101	Diameter of eye section, min.	Level IV	3.2.1	AIE
102	Length of thread section	Level IV	3.2.1	AIE
103	Dimension from bottom thread section to bottom of flange	Level IV	3.2.1	AIE
104	Outside diameter of shank	Level IV	3.2.1	AIE
105	Flange thickness	Level IV	3.2.1	AIE
106	Diameter of flange	Level IV	3.2.1	AIE
107	Diameter of gasket groove	Level IV	3.2.1	AIE
108	Depth of gasket groove, max.	Level IV	3.2.1	AIE
109	True position of gasket groove	Level IV	3.2.1	AIE
110	Perpendicularity of flange	Level IV	3.2.1	AIE
111	Thread pitch diameter, min.	Level IV	3.2.1	AIE
112	Thread major diameter, min.	Level IV	3.2.1	AIE
113	Tension test (see note 1)	5 minimum	3.4	4.4.3.1
114	Ductility test (see note 2)	5 minimum	3.5	4.4.3.2
115	Fracture test (see note 1)	5 minimum	3.6	4.4.3.3
116	Radius at base of shank, max.	Level IV	3.2.1	AIE

1. At least one per cavity.
2. Same samples as Major 113.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.1	TITLE Plug, Universal	SHEET 2 OF 2	DRAWING NUMBER 9347368
			NEXT HIGHER ASSEMBLY 9345325

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
117	Inside diameter	Level IV	3.2.1	AIE
118	True position of shank	Level IV	3.2.1	AIE
119	True position of thread diameter	Level IV	3.2.1	AIE
Minor				
201	Thread pitch diameter, max.	Level II	3.2.1	AIE
202	Width of gasket groove	Level II	3.2.1	AIE
203	True position of flange	Level II	3.2.1	AIE
204	Radius at top of shank, min.	Level II	3.2.1	AIE
205	Height of eye hole	Level II	3.2.1	AIE
206	Width of eye hole	Level II	3.2.1	AIE
207	Angle of groove inside diameter	Level II	3.2.1	AIE
	This space left blank intentionally			
209	Cavity identification missing	Level II	3.2.1	Visual
210	Radii or chamfers missing or incorrect	Level II	3.2.1	Visual
211	Surface finish incorrect	Level II	3.2.1	Visual
212	Evidence of poor workmanship	Level II	3.7	Visual

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.2	TITLE Universal plug assembly (prior to pouring alloy)	SHEET 1 OF 1	DRAWING NUMBER 9345325
			NEXT HIGHER ASSEMBLY

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Primer or sealing compound not present or noted on surface	Level IV	3.2.2	Visual
Minor	None defined			

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.3	TITLE Universal lifting plug assembly (after pouring alloy)	SHEET 1 OF 1	DRAWING NUMBER 9345325 / NEXT HIGHER ASSEMBLY	
CLASSIFICATION	**EXAMINATION OR TEST**	**CONFORMANCE CRITERIA**	**REQUIREMENT PARAGRAPH**	**INSPECTION METHOD REFERENCE**
Critical	None defined			
Major				
101	Salt spray	Note 1	3.2.3	4.4.5
102	Air pressure test	100%	3.2.3	4.4.2
103	Alloy excessively below or excessively above noted surface	Level IV	3.2.3	AIE
Minor				
201	Paint missing, incomplete or on undersurface	Level II	3.2.3	Visual
202	Anti-seize compound missing or incomplete	Level II	3.2.3	Visual
203	Evidence of poor workmanship	Level II	3.7	Visual

1. Sampling plan as specified in TT-C-490.

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.4	TITLE Ternary eutectic alloy	SHEET 1 OF 1	DRAWING NUMBER 9215391
			NEXT HIGHER ASSEMBLY 9345325

CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Melting test	4.4.1	3.2.4	4.4.1
Minor	None defined			

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Conformance Inspection by classification of characteristics

PARAGRAPH 4.3.2.5	TITLE Adhesive	SHEET 1 OF 1	DRAWING NUMBER 9215393 NEXT HIGHER ASSEMBLY	
CLASSIFICATION	EXAMINATION OR TEST	CONFORMANCE CRITERIA	REQUIREMENT PARAGRAPH	INSPECTION METHOD REFERENCE
Critical	None defined			
Major 101	Impact resistance test	4.4.4	3.2.5	4.4.4
Minor	None defined			

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4.4 Testing.

4.4.1 Melting test of ternary eutectic alloy. Five (5) samples of approximately one quarter pound each shall be randomly taken from each lot of ternary eutectic alloy to be used in each lot of fusible plug assemblies. The samples selected shall be individually tested by placing each in a beaker of distilled water at ambient temperature. The temperature of the water shall then be raised at a rate not to exceed 10°F per minute. The water shall be stirred to assure uniform heating. A temperature-indicating device such as a thermometer or thermocouple, suitable for the required temperature range shall be inserted in the water. When the alloy is observed to melt, the indicated temperature shall be recorded. The measured temperature shall then be compared with the value specified on dwg. 9215391 to determine compliance or non-compliance of the sample with the melting point requirement. If any sample fails to comply with the specified melting point, the alloy lot shall be rejected. In addition, all fusible plug assemblies containing the rejected lot of ternary eutectic alloy shall be rejected.

4.4.2 Air pressure test of fusible plug assembly. Every plug assembly shall be subjected to this test. The test shall be performed by sealing the open end of the plug and completely immersing the plug in water. The air pressure specified on dwg. 9345325 shall then be applied through the sealed end of the plug for the specified time intended. Observation shall be made for air bubbles originating from the area of the fusible alloy indicating a leaking plug. All such leaking plugs shall be classed defective and removed from the lot. The plug assemblies shall be thoroughly dried after completion of this test. All equipment used on this test shall be capable of performing the test accurately and consistently. Equipment shall be calibrated prior to the start of each shift. At the start of each shift and every four (4) hours thereafter, the equipment operation shall be verified with a known leaker. If the equipment fails to either calibrate or verify, all plugs tested since the last satisfactory check will be retested

4.4.3 Mechanical properties tests. Five (5) lifting plugs shall be randomly selected from each lot after final heat treatment and prior to the pouring of the alloy for these tests. If any sample part fails to meet any requirement, the lot represented by the failing part shall be rejected.

4.4.3.1 Tension. The lifting plug shall be screwed into a threaded plate or other suitable fixture and a load of 8,000 pounds minimum applied through the medium of a ¾ inch diameter bolt or pin passing through the eye of the plug. The full load shall be transmitted through the screw threads of the plug.

4.4.3.2 Ductility. The plugs used in the tension test shall be cooled to minus 40 degrees Fahrenheit (F.) for a minimum of 30 minutes. The eye of each plug shall be flattened by either

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pressure or blows in any direction until the inner diameter of the eye is permanently reduced by 0.050 inch while at minus 40 degrees F. Each plug shall be visually examined for cracks after flattening.

4.4.3.3 Fracture. The body of each sample iron casting shall be broken through the thickest section for inspection. The plug shall be observed for discontinuous structure, porosity or shrinks.

4.4.4 Impact resistance of adhesive. The bottle of adhesive representing the lot which will be used for a weeks's production will be tested prior to use for impact resistance. A total of ten specimens will be prepared and tested in accordance with the procedures and methods specified in ASTM D-950-54. If any sample tested fails to meet an impact resistance of ten minus one (10-1) foot pounds, the lot or batch of adhesive shall be rejected.

4.4.5 Salt spray. The sampling plan for this test shall be as specified in TT-C-490. The test shall be performed on the plugs as specified in TT-C-490. The number of hours and percentage of sodium chloride solution shall be as indicated in dwg. 9345325. Examination shall be made for any evidence of corrosion of the painted areas. If any plug fails to comply with the specified requirements, it shall be classified defective and corrective action taken as specified in TT-C-490.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The component covered by this specification is intended for use in lifting and handling artillery projectiles. This specification contains requirements suitable for production. There is no known commercial use or application.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. For acquisition purposes, the contract or order should specify packaging requirements. The lifting plugs should be palletized and marked in accordance with ASTM-D-3951 (Standard Practice for Commercial Packaging) and be ordered by the applicable stock number.

c. Requirements for submission of first article sample.

d. Submission of contractor acceptance inspection equipment (AIE)(see 6.3).

e. Provisions for Certification of nondestructive test (NDT) personnel (see 6.4).

f. Establishment of visual defect standards (see 6.5).

g. Submission of ammunition data cards (see 6.6).

6.3 Submission of contractor acceptance inspection equipment (AIE) designs for approval. Submit copies of designs as required to: Commander, U.S. Army TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

6.4 Visual standards. Visual standards for evaluation of metal defects will be established for each contract after the first month of production.

6.5 Submission ammunition data cards. In addition to the normal distribution of records for items procured by the Department of the Army, one copy of all ammunition data cards should be forwarded to Commander, TACOM-ARDEC, ATTN: AMSRD-AAR-QEM-A, and AMSRD-AAR-MEM-C Picatinny Arsenal, NJ 07806-5000.

6.6 Inspection lots. It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of tests or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria per MIL-STD-1168 and sampling provisions of this specification are maintained.

6.7 Visual examination qualification. When compliance with the applicable requirement is in doubt as a result of visual examination the characteristic may be measured or gaged to determine acceptability.

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6.8 Tool Control. Where destruction of components is necessary to inspect these dimensions, measurements of the tool may be substituted provided that the contractor has established a correlation between the tool dimensions and the components dimensions prior to the start of production. The contractor should be responsible for maintaining adequate surveillance to assure the drawing requirements are being met throughout the life of the tools.

6.9 Alternative testing procedures. Alternative testing procedures may be used by the contractor when such testing and procedures provide as a minimum the quality assurance provisions specified herein. Prior to applying the alternative procedures, the contractor should describe them in a written proposal and should demonstrate for the approval of the responsible Government technical agency that their effectiveness is equal to, or better that the quality assurance provisions specified herein.

6.10 Dimensional control of cast or automatic tool parts. In place of the normal sampling associated with the classification of defects, and subject to the approval of the government procuring agency, the optional sampling plans in paragraphs 6.10.1 and 6.10.2 may be used for dimensional control of cast or automatic tool parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die or mold cavity. For all specified test, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die or mold cavity, these optional plans are not applicable. In the event of in-process reject(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, should require approval of the government procuring agency.

6.10.1 Cast parts. A sample of at least five parts (as cast) from each cavity should be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production should be performed on the above quantities from each cavity after the production of 5000 parts. Individual cavity identification must be provided. If any defective parts are found during the qualification of the cavity, the cavity producing the defective parts should not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part should be removed from production. Further, that portion of production since the last control check should be returned for corrective action. All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above.

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6.10.2 Automatic tool parts (see 6.11.b). A sample of at least 10 parts should be dimensionally inspected to qualify a new or adjusted setup for use in production. For multiple simultaneous operations (the same operation on 2 or more parts), at least 5 parts per operation per setup should be inspected. In addition, a random sample of 5 parts per setup or simultaneous operation should be selected from the last portion of each hour's production, for dimensional inspection, as a control of the setup during production. If an hour's production is 250 parts or less per setup or simultaneous operation, the hourly sample may be reduced to 3 parts per setup or simultaneous operation. Automatic tools should be considered for this reduced inspection only when equipped with devices to provide automatic shut-down or clear immediate warning if machine power input (electricity, air or oil pressure) fall below the minimum level necessary for normal machine operation. All parts in process when power input runs below minimum should be inspected for dimensional requirements prior to acceptance. Such parts should not be counted as qualification or control samples. If any defective parts are found during qualification of a setup, that setup should not be used in production. If any defective parts are found when inspection is performed for control of a setup, the setup producing the defective part should be removed from production. Further, that portion of production since the last setup control check should be returned for corrective action. All setups removed from production because of some fault may, after adjustment or re-tooling as necessary, be returned to production providing they pass the qualification test above.

6.11 Definitions

a. Acceptance Inspection Equipment (AIE). Any measuring device that is traceable to the national or international standard which is used to assure conformance of material to the contract requirements.

b. Automatic tool. An automatic tool is any self-regulating piece or assemblage of production equipment which is setup or programmed to perform one or more operation on a part. The tool operator's functions should not exceed part handling and input of a start signal. The tool setup should provide positive, repetitive, three plane positioning of the work piece.

6.12 Amendment notations. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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6.13 Subject term (keyword) listing.
Salt Spray
Ductility
Ternary
Eutectic

Custodian:
Army - AR

Preparing Activity:
Army - AR
(Project 1320-2011-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil.

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